

FERROELECTRIC GATE-ALL-AROUND FETS FOR NEXT-GENERATION NEUROMORPHIC SYSTEMS: DEVICE MODELING AND SPIKING NETWORK IMPLEMENTATION

EEE-4100: Project Work

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ABSTRACT

This work focuses on enabling energy-efficient neuromorphic computing through the design and characterization of a novel, steep-switching Gate-All-Around Ferroelectric FET (GAA-FeFET) using technology computer-aided design (TCAD). The CMOS-compatible device will be engineered to utilize its hysteretic and partial-polarization-switching dynamics to function as a key neuromorphic primitive, the Leaky-Integrate-and-Fire (LIF) spiking neuron. The potential for its use as a multi-level analog synapse will also be investigated. Sub-60 mV/decade switching and non-volatile memory properties will be introduced by incorporating a ferroelectric HfZrO_x (HZO) layer within the GAA gate stack. The first half of the project will consist of designing a nanosheet-based GAA-FeFET structure and calibrating the TCAD ferroelectric models with experimental data reported in the literature. The second half will focus on extracting a compact model of the device's primary neuromorphic characteristic (LIF neuron) and implementing this model within a Python-based Spiking Neural Network (SNN). This SNN will be designed for a clinically relevant application involving the classification of cardiac arrhythmias from the MIT-BIH ECG database, using a system-level approach for evaluating neuromorphic device performance. Data obtained from the device and system-level simulations will be thoroughly analyzed to evaluate the proposed GAA-FeFET's performance as a scalable neuromorphic hardware primitive.

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CHAPTER 1

INTRODUCTION

I. BACKGROUND AND CONTEXT:

The computational paradigm of modern electronics is built upon the von Neumann architecture, which establishes a physical separation between processing (CPU) and memory (RAM) units. For decades, this model has delivered exponential performance gains via Moore's Law. However, for data-intensive tasks such as artificial intelligence (AI) and real-time signal processing, this architecture faces a critical limitation known as the "von Neumann bottleneck" [1]. The constant shuffling of data between the processor and memory consumes the majority of system energy and time, creating an efficiency barrier that cannot be solved by simple transistor scaling.

To overcome this bottleneck, the field of neuromorphic computing has emerged, drawing inspiration from the structure and efficiency of the biological brain. The brain performs "in-memory computing" by co-locating processing and memory within dense networks of neurons and synapses. Spiking Neural Networks (SNNs), which operate using asynchronous spikes rather than continuous signals, emulate this biological behaviour and offer substantial gains in energy efficiency [2]. Neuromorphic computing has also begun to influence artificial intelligence, enabling AI systems to process data more efficiently and perform complex tasks at low power, particularly in edge and real-time applications [3].

To fully realize the potential of SNNs, new hardware is required. Ferroelectric Field-Effect Transistors (FeFETs) have become a leading candidate for this hardware. Based on CMOS-compatible materials like Hafnium Zirconium Oxide ($HfZrO_x$ or HZO), FeFETs exhibit non-volatile memory and rich polarization-switching dynamics [4] [5]. properties allow a single FeFET to emulate the behavior of both a spiking neuron and an analog synapse, offering a path to dense, low-power neuromorphic hardware. This biological process, which is the blueprint for neuromorphic computing, is illustrated in Figure 1, showing how input spikes are processed through synapses and neurons to produce an integrated 'fire' response.

Simultaneously, transistor architecture is evolving from FinFETs to Gate-All-Around (GAA) structures. GAA transistors, which wrap the gate around the channel on all sides,

provide superior electrostatic control, reduced short-channel effects, and higher drive currents [6].

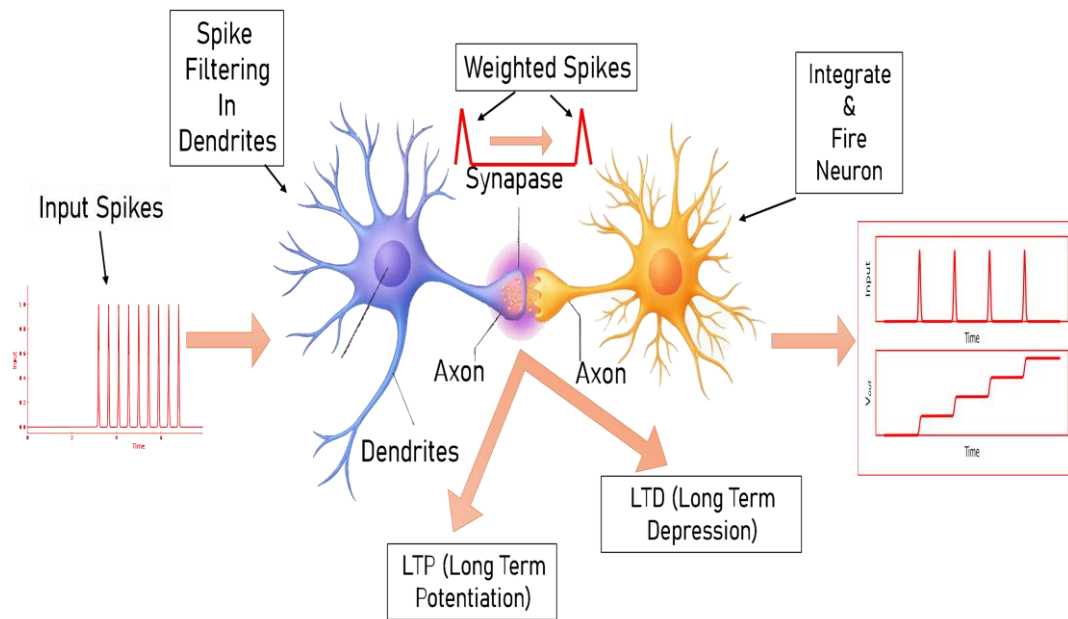


Figure 1: Biological blueprint of Neuromorphic Computing

The advantages of using a FeFET-based approach for neuromorphic computing are manifold. Firstly, its intrinsic non-volatility allows for zero standby power, crucial for edge devices [7]. Secondly, its compatibility with existing CMOS fabrication processes makes it commercially scalable, unlike more exotic materials [5]. Thirdly, the device physics are well-suited for neuromorphic functions: the hysteretic switching is ideal for creating relaxation oscillators (neurons), and the partial polarization switching allows for a high number of analog conductance states (synapses) [8]. This project will harness these advantages in a state-of-the-art GAA structure. Figure 2 summarizes this concept, outlining how the superior electrostatic control of a GAA structure and the non-volatile polarization of ferroelectric HZO combine to enable high-density, energy-efficient neuromorphic computing.

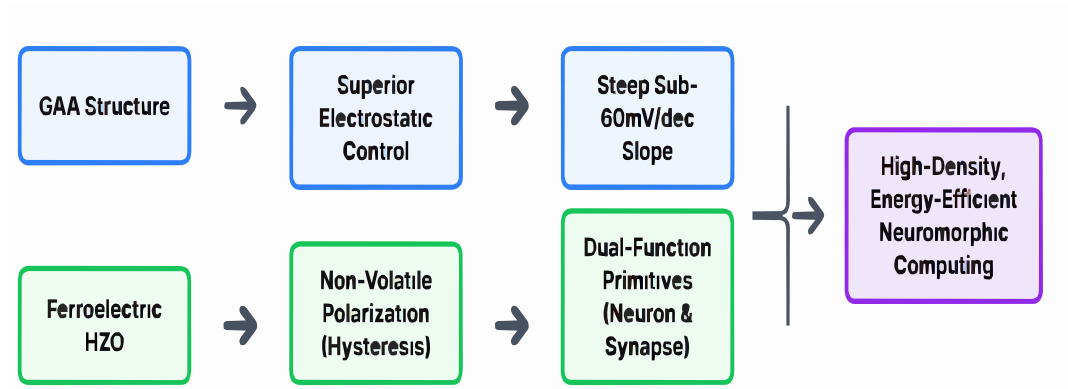


Figure 2: Core Advantages of a GAA-FeFET Neuromorphic Platform.

II. OBJECTIVES:

Development of the Transistor Architecture:

At the beginning of our project, we focus on designing a novel Gate-All-Around (GAA) nanosheet Ferroelectric FET (FeFET) using Sentaurus TCAD. The transistor structure will be designed and calibrated with attention to the nanosheet dimensions, including thickness and width, as well as the ferroelectric gate stack thickness (HfZrO_x). The goal is to achieve a device that demonstrates stable hysteresis and a steep subthreshold slope below 60 mV/dec, enabling efficient low-power operation.

Analysis of Neuromorphic Functionality:

The main aim of our project is to analyze the neuromorphic computing behaviors of the GAA-FeFET. This includes studying the curves obtained from Sentaurus simulations to understand the device's response. The transistor will be modeled as a Leaky-Integrate-and-Fire (LIF) neuron in a relaxation oscillator circuit to examine its spiking dynamics. In addition, the device will be evaluated as an analog synapse by applying potentiation and depression pulses, allowing us to quantify its multi-level conductance states.

System-Level Integration for Neuromorphic Tasks:

This project also seeks to demonstrate the integration of the GAA-FeFET into a system-level neuromorphic application. A Spiking Neural Network (SNN) for ECG arrhythmia classification will be developed in Python, inspired by the work of Yuan et al. on VO_2 memristive devices [9]. This will involve creating a spike encoder for the MIT-BIH

dataset and designing a custom SNN layer in PyTorch that incorporates the characteristics of the simulated FeFET in place of conventional neuron models. The network will be trained using the Surrogate Gradient method and benchmarked against existing literature to evaluate classification performance. The overall workflow of these objectives, from device design to system-level implementation and evaluation, is depicted in the flowchart in Figure 3.

The primary objectives for the first semester centered on establishing the foundational simulation framework necessary for the GAA-FeFET design. This involved two main goals. First, to construct the device architecture for a 2D back-gated MoS₂ Negative Capacitance FET (NC-FET) within the Sentaurus Structure Editor (SDE). This specific device was chosen to serve as the initial platform for calibration, as it directly corresponds to experimental data available in the literature. Second, to initiate the model calibration process by simulating the device's I_d - V_g transfer characteristics and comparing these initial results against the published experimental data. The aim was to validate the baseline TCAD setup and identify key parameters, such as the LK coefficients, requiring rigorous tuning.

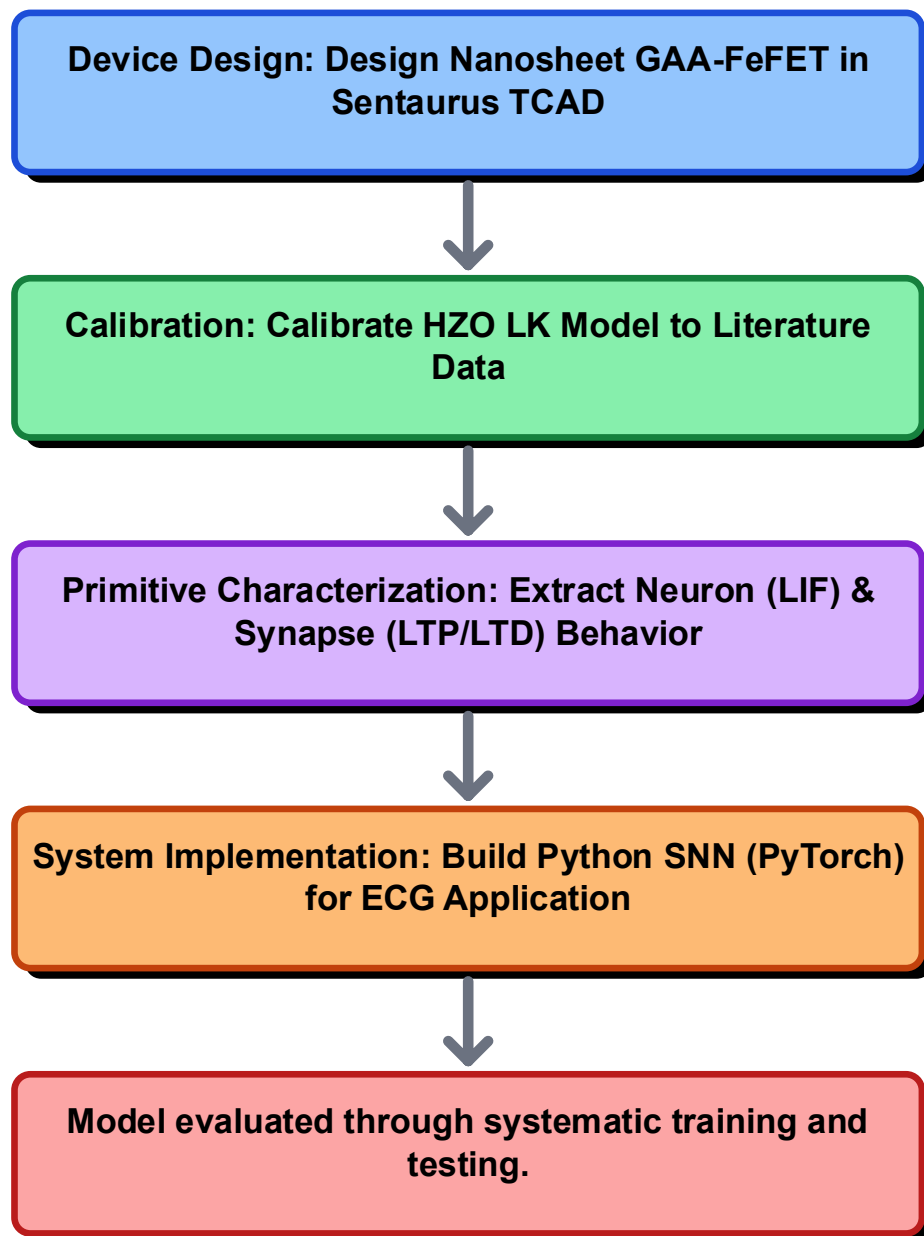


Figure 3: Objectives of the project

CHAPTER 2

LITERATURE REVIEW

Efforts to continually scale down traditional semiconductor technologies have begun to run into inherent physical limitations, often described as the Von Neumann bottleneck. In this conventional architecture, computing units (logic cores) and data storage elements (memory blocks) are physically separated, which leads to inefficiencies in power usage, increased latency, and heat generation when handling the large volumes of data typical in modern applications. To address these challenges, neuromorphic computing has emerged, aiming to replicate the human brain's parallel, event-driven, and energy-efficient processing, where logic and memory functions are closely integrated. Central to this approach are synaptic devices, which function as analogue weight elements. Among these, the Ferroelectric Field-Effect Transistor (FeFET) stands out as a strong candidate due to its CMOS compatibility, non-volatile behavior, compact footprint, and very low write energy. The Gate-All-Around FeFET (GAA-FeFET) further enhances device performance by offering superior electrostatic control, reduced short-channel effects, and improved scalability.

Advanced semiconductor device architectures, such as the Gate-All-Around (GAA) transistor, are crucial for continuing scaling efforts beyond current limitations. The GAA structure allows for the aggressive downscaling of transistors and is a key technology for enhancing efficiency in neuromorphic applications. N. Loubet et al. demonstrated the fundamental capability of stacked nanosheet GAA transistors to enable scaling beyond the performance limits of the FinFET architecture. The shift toward GAA structures addresses previous challenges in conventional FET-based leaky integrate-and-fire (LIF) neurons, particularly the need for a floating body and the constraints on scaling the silicon-on-insulator (SOI) film in the sub-nanometer range [10]. Y. Bhatawdekar et al. utilized the GAA Floating Nanosheet Field-Effect Transistor (FNSFET) structure to design an energy-efficient LIF neuron, capitalizing on the enhanced control over short-channel effects inherent to the GAA geometry. The proposed GAA FNSFET LIF neuron design, featuring a gate length (L_g) of 100 nm, achieved an exceptionally low energy consumption of 4.88 fJ (4.88×10^{-15} J) per spike, operating at a supply voltage of 1 V [11].

The incorporation of ferroelectric (FE) materials, such as HfO_2 based oxides, into the gate stack has enabled the exploration of Negative Capacitance (NC) field-effect transistors (NCFETs). NCFETs utilize the intrinsic negative slope of the ferroelectric material's capacitance-voltage (C-V) curve to amplify the internal voltage, potentially lowering the subthreshold swing (SS) below the classical 60 mV/decade limit, which is essential for ultra-low power applications. M.-H. Lee et al. reported experimental results for ferroelectric HfZrO_x FETs demonstrating the benefits of this material system for NCFETs. Their optimized devices, showing a CET = 0.98 nm, achieved an impressive minimum reverse subthreshold swing (SS_{rev}) of 28 mV/dec and a forward SS (SS_{for}) of 42 mV/dec, validating the potential for steep switching [12]. M.-Y. Kao investigated the impact of the material microstructure on device performance, particularly addressing variability arising from the non-uniform distribution of dielectric (DE) and ferroelectric (FE) grains within the gate stack. Through TCAD simulations, Kao quantified that the random spatial distribution of DE and FE phases resulted in variations of the ON and OFF currents up to 14.44% and 30.23%, respectively. This highlights that while NCFETs offer performance benefits, controlling microstructural non-uniformity is crucial for minimizing variability in scaled devices [13].

Ferroelectric FETs (FeFETs) are highly promising candidates for implementing energy-efficient, non-volatile analog synapses in neuromorphic hardware due to their CMOS compatibility and the ability of their ferroelectric layer to store multiple conductance states. M. Jerry et al. successfully demonstrated the use of a FeFET as an analog synapse for accelerating deep neural network (DNN) training. Their work showed that a FeFET could operate as a multi-state weight cell, achieving $4 \times$ conductance modulation and requiring approximately 75 ns program pulses. This multi-bit operation is achieved through the progressive switching of ferroelectric domains using sub-coercive voltage pulses [14]. J. Zhao et al. optimized the operational bias schemes for HfAlO_x (HAO)-FeFET synapses in neural network (NN) systems, focusing on improving weights linearity, accuracy, and energy efficiency during online training. Using measured data from HAO-FeFETs under incremental pulses, they achieved a minimum nonlinearity of 0.01 for both potentiation (increasing conductance) and depression (decreasing conductance), demonstrating high suitability for online training [15].

The hysteretic switching characteristics of FeFETs make them excellent candidates for building compact, capacitor-less artificial spiking neurons that mimic the Leaky Integrate-and-Fire (LIF) dynamics of biological neurons. The dynamics are typically realized through relaxation oscillators utilizing the polarization switching properties. Z. Wang et al. experimentally demonstrated the fundamental operation of ferroelectric spiking neurons based on the FeFET structure for applications such as unsupervised clustering. This approach permits the realization of a FeFET-based LIF neuron using only seven transistors (including one FeFET), with an estimated footprint area of approximately $1.74 \times 1.18 \mu\text{m}^2$ at the 45nm technology node [16]. C. Jiang et al. analyzed the crucial design parameters that govern the hysteretic characteristics required for oscillation. Their analytical models showed that the critical thickness (tFE, cri) necessary to induce hysteresis in a FeFET was calculated to be 31.2 nm for a drain-source voltage (V_{ds}) of 0.05 V and 43.75 nm for $V_{\text{ds}} = 1 \text{ V}$ [17].

In the context of neuromorphic applications, several research efforts have explored the integration of ferroelectric and advanced transistor technologies within spiking neural network (SNN) frameworks. M. A. Khanday et al. proposed a bio-inspired spiking neuron based on a double-gate ferroelectric tunnel FET (DG-FE-TFET) and implemented it within a multi-layer spiking neural network (SNN) for heartbeat classification. Utilizing heartbeat signals derived from the MIT-BIH database, the system achieved classification of heartbeats into four target classes (N, F, SVEB, and VEB). Leveraging the device's tunneling mechanism for low voltage operation, the proposed 50 nm DG-FE-TFET neuron consumes ultra-low energy of 0.58 aJ per spike ($0.58 \times 10^{-18} \text{ J}$) and operates at a high spiking frequency of 344 GHz. The three-layer SNN, implemented and simulated in Python, achieved a maximum classification accuracy of 84.4% [18]. Y. He et al. demonstrated a spiking neural network (SNN) utilizing HfO₂-based Ferroelectric Tunnel Junction (FTJ) memristors as synapses for a handwritten digit recognition task. Using a pure SPICE network simulation with the FTJ compact model, they achieved a successful recognition rate of 78.8% when testing with 1000 samples, demonstrating the application of ferroelectric devices in complex neuromorphic systems [19]. J.-K. Han et al. successfully demonstrated high-scalability artificial neurons (vertical Si Nanowire GAA 1T-neurons) that were applied in a two-layer fully connected SNN for handwritten digit recognition using the MNIST dataset.

Utilizing the measured neuronal characteristics in a software simulation, they reported a high recognition accuracy of 93% [20].

Effective training of SNNs requires specialized algorithms compatible with spike-based event processing and the non-volatile nature of the underlying hardware (e.g., FeFETs or FTJs). Y. He et al. relied on the fundamental principles of Spike-Timing-Dependent Plasticity (STDP) to train the FTJ-based synaptic weights in their handwritten digit recognition system. The training process involves modifying the conductance (synaptic weight) based on the relative timing of presynaptic and postsynaptic spikes, driving the continuous evolution of the FTJ array conductance [19]. S. Dutta et al. proposed using the Surrogate Gradient (SG) learning algorithm to enable supervised learning on their all-FeFET-based SNN platform. Their hardware implementation utilized the rich stochastic dynamics of the FeFET neuron, demonstrating spike frequency adaptation where the instantaneous firing rate of the neuron decreases with each output spike which is a key biological feature important for homeostatic plasticity [21].

CHAPTER 3

PROPOSED METHODOLOGY

I. Project Design:

- **Overall Design:**

The project is organized following a device-to-system workflow and is divided into into three interconnected stages. Each stage builds upon the outcomes of the previous one, creating a smooth progression from device-level design to system-level application.

Phase 1: Designing the Transistor Architecture: The first stage involves designing a novel GAA-FeFET nanosheet in Sentaurus TCAD. During this process, the device's physical models will be carefully calibrated to ensure accurate representation of its electrical behavior.

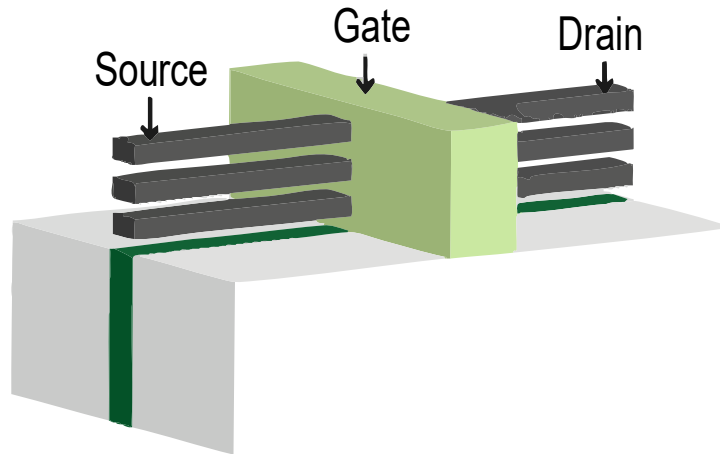


Figure 4: 3D view of Gate All Around Transistor

As shown in Figure 4, the three-dimensional view illustrates the structural configuration of the Gate-All-Around (GAA) transistor. The device will employ a nanosheet-based structure, chosen over nanowires for its improved electrostatic control and higher drive current, consisting of silicon nanosheets fully wrapped by a metal-ferroelectric-insulator-semiconductor (MFIS) gate stack. The channel will use silicon nanosheets, while the gate stack will comprise a TiN metal gate, a $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ (HZO) ferroelectric layer, and a thin SiO_N or SiO_2 interfacial layer. The 1:1 Hf:Zr ratio ensures a robust ferroelectric phase suitable for memory and switching applications. The simulation will

integrate standard drift-diffusion transport models with the Landau-Khalatnikov (LK) model for the HZO layer, with LK parameters calibrated to replicate experimental P-E and C-V curves from the literature. The simulation plan begins with ferroelectric calibration using a simple TiN/HZO/TiN capacitor, tuning the LK parameters until the simulated polarization-electric field (P-E) loop aligns with experimental data. After calibration, the complete GAA-FeFET will be simulated to extract its $I_d - V_g$ transfer characteristics in both forward and reverse sweeps, targeting a clear hysteretic memory window and sub-60 mV/dec switching slope indicative of negative capacitance behaviour. The target $I_d - V_g$ transfer characteristic for the GAA-FeFET, demonstrating a sharp turn-on and low off-state current, is shown in Figure 5.

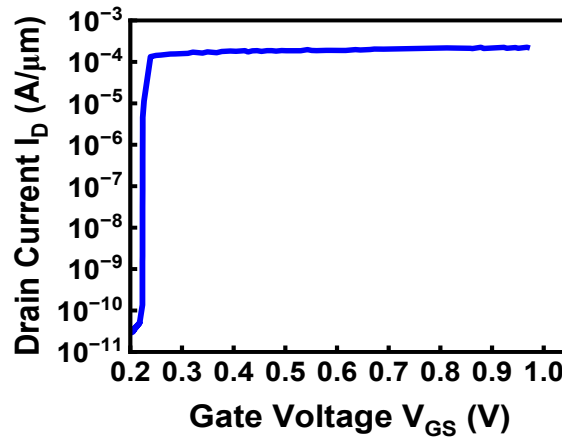


Figure 5: Target $I_d - V_g$ characteristics of a FeFET [17]

Phase 2: Neuromorphic Primitive Characterization: In the second stage, the electrical characteristics obtained from the simulations will be analyzed to model the device as a Leaky-Integrate-and-Fire (LIF) neuron and as an analog synapse, enabling exploration of its neuromorphic functionalities.

Since full TCAD simulations are computationally intensive, a compact model of the GAA-FeFET will be extracted and implemented in Python or SPICE for system-level analysis. The primary aim of this phase is to characterize the device as a neuromorphic primitive, focusing on its behaviour as a Leaky-Integrate-and-Fire (LIF) neuron.

For neuron characterization, a 1FeFET-1T1C relaxation oscillator circuit will be simulated in Python using the compact device model. A constant input current,

representing synaptic activity, will be applied to the circuit, and the capacitor voltage (V_s) will be monitored. The voltage is expected to rise linearly until the FeFET switches ON at a threshold voltage, then discharge abruptly, with intrinsic hysteresis causing a reset at a lower voltage, effectively emulating LIF neuron dynamics.

Phase 3: System-Level Implementation: The final stage focuses on integrating the characterized device into a Python-based Spiking Neural Network (SNN). This network will utilize the device model from Phase 2 and will be applied to a neuromorphic ECG classification task, demonstrating the practical system-level potential of the GAA-FeFET.

In this final phase, the project demonstrates the neuromorphic computing capabilities of the GAA-FeFET by implementing a complete ECG classification system in Python. The MIT-BIH Arrhythmia dataset will be pre-processed into sparse spike trains using a Python-based asynchronous spike encoder, which converts the analog ECG signals into UP, DOWN, and CUE spike channels. A Long Short-Term Memory Spiking Neural Network (LSNN) will be constructed in PyTorch, incorporating a custom GAALIFNeuron layer whose parameters are derived directly from the f-I curve and other device characteristics obtained in Phase 2. The network will be trained using Backpropagation Through Time with the Surrogate Gradient method to handle spiking non-linearities. Performance will be evaluated on a test set from the MIT-BIH database, generating accuracy, sensitivity, and specificity metrics. These results highlight the potential of the proposed GAA-FeFET for practical neuromorphic computing applications. Figure 6 provides a visual diagram of this entire system-level workflow, from the analog ECG signal input to the final classification output.

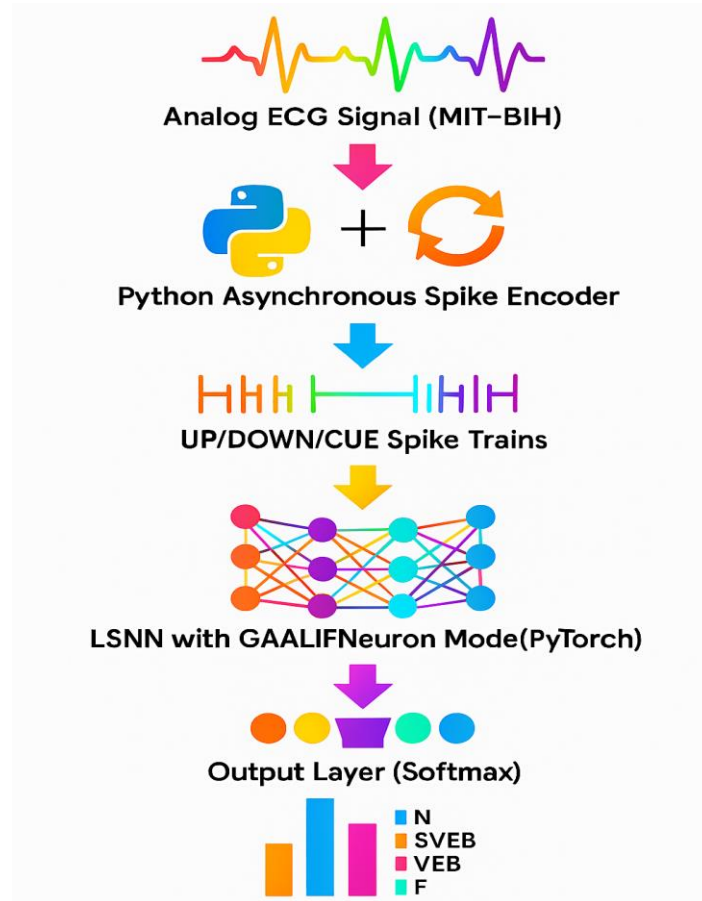


Figure 6: System-Level Diagram for Python SNN-based ECG Classification

The project is expected to be successfully implemented, demonstrating both the transistor-level performance of the GAA-FeFET and its integration into neuromorphic circuits for system-level tasks. The results will provide comprehensive insights into the device's electrical behavior, compact modeling, and its application within a spiking neural network. Figure 7 illustrates the complete workflow that has been described so far, capturing all the stages from device design to system-level implementation.

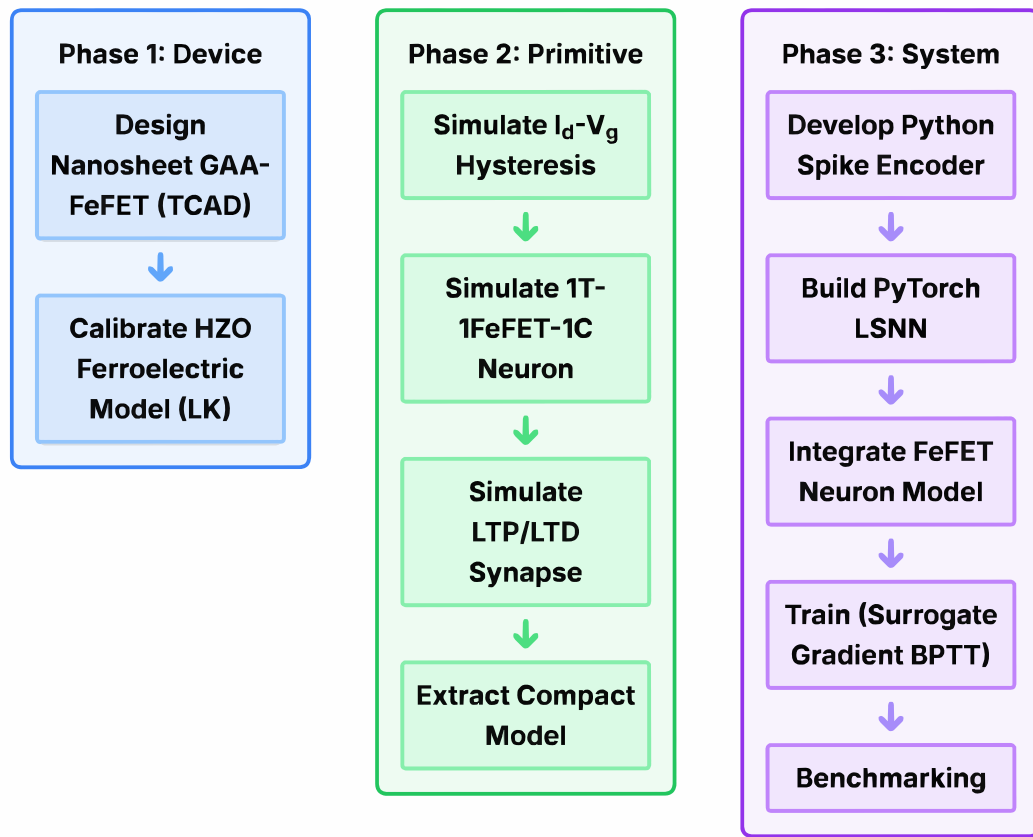


Figure 7: Overall Project Workflow

- **Work done during 1st semester**

During the first semester, the project focused on establishing the TCAD simulation framework. The main goal was to construct the device architecture in Sentaurus and initiate the model calibration process, which is essential for simulating the proposed GAA-FeFET.

As part of this effort, a 2D back-gated MoS₂ Negative Capacitance FET (NC-FET) was designed in the Sentaurus Structure Editor (SDE). This structure was selected as the initial calibration platform because it directly corresponds to the experimental results reported by Jiang et al. [3], providing a reliable reference for tuning the simulation parameters.

The next step involved calibrating the TCAD physics models, particularly the Landau-Khalatnikov (LK) model for the HZO layer, against the experimental data. An initial

simulation of the $I_d - V_g$ transfer characteristics was carried out using the LK coefficients and device parameters from the literature. However, the first-pass simulation showed significant deviations from the experimental curves, with discrepancies in threshold voltage and subthreshold slope, indicating that the published parameters alone were insufficient to reproduce the steep-switching behaviour ($SS_{rev} = 52.3\text{mV/dec}$) [3]. Figure 8 shows the structure of our simulated device, which is a Metal-Ferroelectric-Metal-Insulator-Semiconductor (MFMIS) Device.

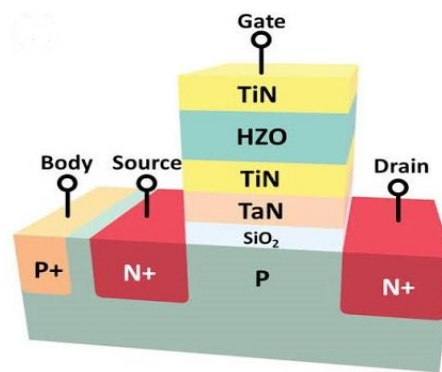


Figure 8: Sentaurus SDE model of the MFMIS Device [17]

In conclusion, the 1st-semester work successfully set up the TCAD structural framework in SDE, but the initial calibration highlighted the need for a more systematic tuning of the LK parameters and other key physical properties, such as interfacial layers and contact resistances. This finding establishes the immediate priority for the second semester, which is to perform a rigorous parameter calibration so that the NC-FET model can serve as a validated baseline for the eventual GAA-FeFET design.

- **Detail work plan for 2nd semester**

In the second semester, the project will build upon the foundational work from the first semester to complete a full device-to-system study. The initial focus will be on rigorous calibration and validation of the NC-FET model. TCAD physics parameters, including Landau coefficients, interface traps, and contact resistances, will be systematically tuned to align the simulated $I_d - V_g$ and P-E curves with experimental data. Once

validated, the model will serve as the basis for designing the nanosheet GAA-FeFET in Sentaurus, where baseline $I_d - V_g$ and C-V simulations will confirm proper hysteresis, memory window, and subthreshold slope behavior.

Following the device design, a compact model of the GAA-FeFET will be developed in Python for neuromorphic system simulations. This model will be used to simulate a 1FeFET-1T1C relaxation oscillator circuit, allowing detailed characterization of its LIF neuron properties and the generation of the frequency-current curve. Additional neuromorphic features, such as preliminary synaptic behaviour including long-term potentiation (LTP) and depression (LTD), will also be simulated if time permits, providing insight into the device's analog state precision and bit-level resolution for neuromorphic applications.

Finally, the focus will shift to full neuromorphic system implementation in Python. The MIT-BIH Arrhythmia dataset will be pre-processed using a Python-based asynchronous spike encoder, converting analog ECG signals into sparse UP, DOWN, and CUE spike trains suitable for spiking neural networks. A Long Short-Term Memory Spiking Neural Network will be constructed in PyTorch, incorporating a custom GAALIFNeuron layer whose parameters are directly derived from the compact device simulations. The network will be trained using Backpropagation Through Time with the Surrogate Gradient method, and its performance as a neuromorphic computing system will be evaluated in terms of accuracy, sensitivity, and specificity.

- **Expected Outcome**

This project will deliver a comprehensive device-to-system study of a novel neuromorphic computing element. The first outcome is a fully calibrated TCAD simulation model of a nanosheet GAA-FeFET, which captures the intricate ferroelectric and transport physics of the device and serves as a reliable foundation for designing brain-inspired circuits. The second outcome is a detailed quantitative characterization of the GAA-FeFET's performance, demonstrating its ability to function both as an LIF spiking neuron through the frequency-current curve and as an analog synapse via LTP and LTD behavior with defined bit precision, effectively replicating key aspects of neuronal and synaptic activity. The final outcome is a working Python-based Spiking custom neuron model is physically grounded in the TCAD simulations. This

demonstrates the potential of the device to support successful neuromorphic applications and highlights its relevance for biomedical signal processing and other brain-inspired computing tasks that require energy-efficient, parallel, and event-driven computation.

II. Gantt Chart:

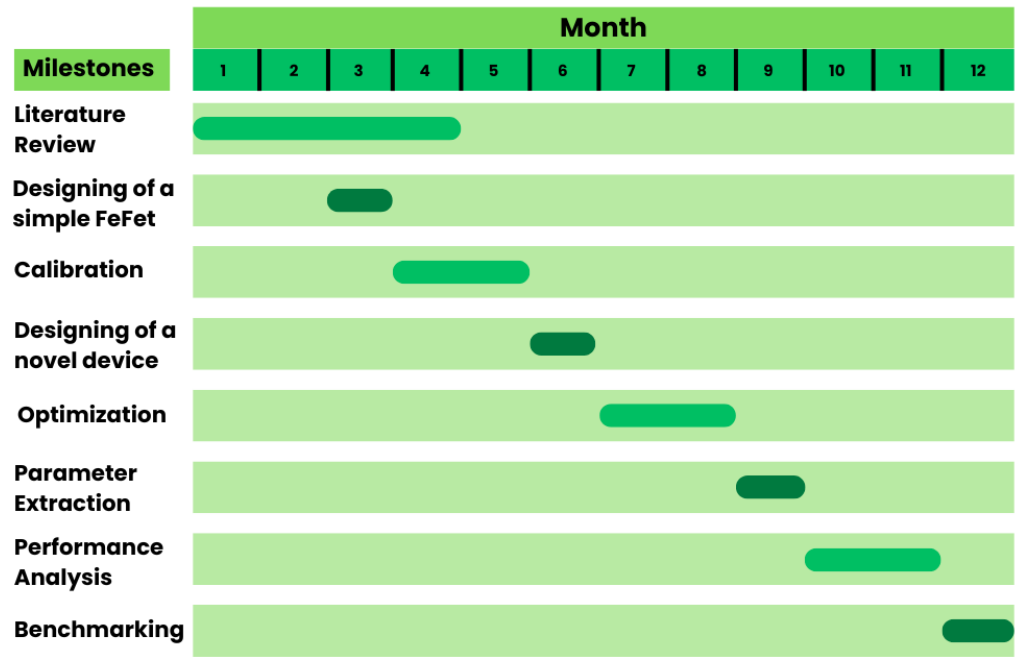


Figure 9: Gantt Chart of project workflow

CHAPTER 4

CONCLUSION

To date, we have established a complete TCAD framework for simulating and analyzing ferroelectric transistors, which now serves as the foundation for developing a highly efficient GAA-FeFET aimed at enabling advanced neuromorphic functionalities. This phase of the work has provided valuable insight into the complex interaction between ferroelectric polarization, interface properties, and device electrostatics. The next step involves fine-tuning these models to achieve accurate calibration and to capture the steep-switching and memory behavior required for neuromorphic operation. In parallel, efforts are being directed toward developing compact models that can replicate the observed device characteristics at the circuit and system levels. These models will be essential for simulating neuromorphic behaviors such as spike generation and synaptic plasticity. One of the key challenges lies in achieving precise parameter calibration across different simulation environments while establishing a stable and compatible Python setup that can seamlessly integrate the calibrated TCAD data for system-level analysis. Work is also underway to link the device-level results to a functional spiking neural network model in Python, enabling a direct evaluation of the GAA-FeFET's potential in biomedical signal processing applications.

CHAPTER 5

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